BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 8

Rational Functions and Partial Fractions

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1. INTRODUCTION

In Unit 7, we delved deep into several types of special functions that are pivotal in higher mathematics and have broad applications across various fields. We explored logarithmic functions, uncovering their fundamental definitions, key properties, and the principles underlying their derivatives. This set the foundation for understanding the exponential growth and decay models frequently encountered in finance and natural sciences.

We then transitioned to hyperbolic functions—functions that mirror the familiar trigonometric functions but are related to hyperbolas instead of circles. Their unique characteristics, such as sinh, cosh, and tanh, alongside their graphs and identities, were discussed in detail, including the inverse hyperbolic functions which are crucial for solving certain types of equations in calculus.

The study of inverse circular or trigonometric functions rounded out the unit. These functions are essential for calculating angles from given trigonometric ratios, providing insights into the behavior of waves, oscillations, and circles, with their principal value branches ensuring uniqueness of the resulting angle—a concept critical for clear geometric interpretations and practical applications in engineering and physics.

As we progress to Unit 8, we focus on "Rational Functions and Partial Fractions", a key area in algebra and calculus that deals with ratios of polynomials. For students in computer applications, understanding these functions is crucial as they often model scenarios where rates of change are expressed as ratios, such as in rendering algorithms or in computational simulations of natural phenomena.

Rational functions can be complex, characterized by the behavior of their graphs such as asymptotes and discontinuities, which are vital for understanding the limits and behavior of functions as inputs grow large or small. The decomposition of these functions into partial fractions simplifies the integration and transformation of complex rational expressions into simpler, more manageable terms. This simplification is particularly beneficial in algorithm design and error analysis in numerical methods, where breaking down complex problems into simpler sub-problems is a common strategy.

Studying these advanced concepts requires a structured approach. Begin with a thorough review of polynomial functions and their properties, as these are the building blocks of rational functions. Engage with interactive tools and graphing calculators to visualize these functions for a better conceptual understanding. Practice decomposing complex rational expressions both by hand and using software tools, which is essential for developing a deeper insight into the function behaviours and their applications.

Focus on applying these concepts through practical problems and case studies, especially those that simulate real-world scenarios relevant to computer applications. This practical application solidifies theoretical knowledge and enhances problem-solving skills, preparing students for both academic and professional challenges in computational fields. By the end of this unit, you should be able to not only solve complex equations involving rational functions but also apply these skills in optimizing and improving software applications that require high levels of mathematical calculations.

1.1. Learning Objectives

At the end of this unit, you will be able to:

- *Define the key terms related to rational functions including "proper rational function", "improper rational function", and "partial fractions".*
- *Explain the process of decomposing a rational function into partial fractions and the significance of each step in the decomposition process.*
- *Perform the decomposition of various rational functions into partial fractions using standard algebraic techniques.*
- *Distinguish between different types of rational functions based on the degrees of their numerator and denominator polynomials and decide the appropriate method for their decomposition.*
- *Assess the accuracy of partial fraction decompositions by verifying the consistency of results through back-substitution into the original rational function.*
- *Formulate complex rational functions and develop strategies for their decomposition into simpler fractions that can be easily integrated or used in further calculations.*

2. PARTIAL FRACTIONS

Partial fractions refer to the technique of expressing a complicated rational function as a sum of simpler fractions. This method is particularly useful in calculus for integrating rational functions, as well as in algebra for solving equations involving polynomials. By breaking down a complex fraction into more manageable components, partial fractions simplify the process of performing algebraic manipulations and integrations.

2.1. Methods to Solve Partial Fractions

The approach to solving partial fractions varies depending on whether the rational function is proper or improper, and on the factorization of the denominator.

2.1.1. Polynomial of Degree n

A polynomial P(x) of degree n is an expression of the form:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

Where $a_n, a_{n-1}, \ldots, a_0$ are constants, and $a_n \neq 0$. The degree o the polynomial is the highest power of x that appears in the Polynomial with a non-zero coefficient.

2.1.2 Rational Function (Polynomial Fraction)

A rational function is a fraction of two polynomials, expressed as:

$$\frac{P(x)}{Q(x)}$$

Where P(x) and Q(x) are polynomials. The degree of P(X) and Q(x) determines whether the rational function is proper or improper.

2.2. Proper Rational Function

A proper rational function is one where the degree of the numerator polynomial P(x) is less than the degree of the denominator polynomial Q(x). That is:

$$deg\big(P(x)\big) < deg\big(Q(x)\big)$$

2.3. Improper Rational Function

An improper rational function is one where the degree of the numerator polynomial P(x) is greater than or equal to the degree of the denominator polynomial Q(x). That is:

$$deg\big(P(x)\big) \geq deg\big(Q(x)\big)$$

For improper rational functions, polynomial division is performed first to convert it into a polynomial plus a proper rational function. This step ensures that the rational function can be decomposed into simpler parts using partial fractions.

2.4. Decomposing Proper Rational Functions

The decomposition of a proper rational function depends on the factorization of the denominator Q(x). Here are the main methods:

2.4.1 Distinct Linear Factors

If Q(x) can be factored into distinct linear factors, the decomposition takes the form:

$$\frac{P(x)}{(x-r_1)(x-r_2)-(x-r_k)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \cdots + \frac{A_k}{x-r_k}$$

Where $A_1, A_2, \cdots A_k$are constants to be determined.

2.4.2 Repeated Linear Factors

If Q(x) has repeated linear factors, the decomposition is:

$$\frac{P(x)}{(x-r)^n} = \frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_n}{(x-r)^n}$$

2.4.3 Irreducible Quadratic Factors:

If Q(x) has irreducible quadratic factors (quadratic polynomials that cannot be factored further over the real numbers), the decomposition includes terms of the form:

$$\frac{P(x)}{(x^2+bx+C)} = \frac{Ax+B}{x^2+bx+C}$$

2.5. Steps for Partial Fraction Decomposition

1. **Factor the Denominator:** Start by factoring the denominator Q(x) completely into linear and/or irreducible quadratic factors.
2. **Set Up Partial Fractions**: Based on the types of factors, set up the partial fraction decomposition by assigning constants A,B, to the numerators of the partial fractions.
3. **Determine Constants:** Multiply both sides of the equation by the denominator to clear the fractions, then solve for the unknown constants by equating coefficients of corresponding powers of x or by substituting suitable values of x.

4. **Integrate (if applicable):** Use the partial fractions to perform the integration or solve the problem at hand.

Example

Decomposing a Proper Rational Function

$$\frac{3x+5}{(x-1)(x+2)}$$

Solution:

Factor the denominator:

$$(x-1)(x+2)$$

Set up partial fractions:

$$\frac{3x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

Clear fractions by multiplying by the denominator:

$$3x+5 = A(x+2) + B(x-1)$$

Solve for A and B by substituting convenient values for x:

Let

$$x = 1: 3(1) + 5 = A(1+2) + B(1-1) \Rightarrow 8 = 3A \Rightarrow A = \frac{8}{3}$$

Let

$$x = -2: 3(-2) + 5 = A(-2+2) + B(-2-1) \Rightarrow -1 = -3B \Rightarrow B = \frac{1}{3}$$

So, the partial fraction decomposition is:

$$\frac{3x+5}{(x-1)(x+2)} = \frac{8/3}{x-1} + \frac{1/3}{x+2}$$

Partial fractions are a crucial mathematical tool for simplifying and solving problems involving rational functions. Understanding the distinction between proper and improper rational functions, as well as the methods for decomposing them, is essential for effectively utilizing this technique in algebra and calculus. With this theoretical foundation, one can confidently approach a wide range of problems involving partial fractions.

3. SIMPLE LINEAR FACTORS

Simple linear factors are first-degree polynomials of the form (x-r) where r is a constant. In a rational function, if the denominator can be factored into distinct linear factors, each of these factors contributes to the partial fraction decomposition.

Decomposition Method

For a rational function where the denominator consists of distinct linear factors, the decomposition is straightforward. If the rational function is:

$$\frac{P(x)}{(x-r_1)(x-r_2)\cdots(x-r_k)}$$

Then the partial fraction decomposition will be:

$$\frac{P(x)}{(x-r_1)(x-r_2)-(x-r_k)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \cdots + \frac{A_k}{x-r_k}$$

Where $A_1, A_2, \cdots A_k$are constants to be determined.

Example:

As seen in the previous section.

4. REPEATED LINEAR FACTORS

Repeated linear factors occur when the denominator of a rational function has factors of the form $(x-r)^n$ where $n > 1$. These factors require a different approach to decomposition since each power of the repeated factor must be accounted for in the partial fraction decomposition.

Decomposition Method

For a rational function with repeated linear factors, the decomposition includes terms for each power of the repeated factor. If the denominator has a repeated factor $(x-r)^n$ the decomposition is:

$$\frac{P(x)}{(x-r)^n} = \frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \cdots \frac{A_n}{(x-r)^n}$$

Example: Decompose $\frac{3x+5}{(x-1)^2}$

Solution

Set up the Partial Fractions:

$$\frac{3x+5}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$$

Clear the fractions by multiplying through by the denominator

$$3x+5 = A(x-1) + B$$

Solve for A and B by substituting convenient values of x:

Let

$$x = 1: 3(1) + 5 = B \Rightarrow B = 8$$

Differentiate both sides with respect to x and then substitute x = 1:

$$A = 3$$

Therefore, the decomposition is;

$$\frac{3x+5}{(x-1)^2} = \frac{3}{x-1} + \frac{8}{(x-1)^2}$$

5. QUADRATIC FACTORS

Quadratic factors are second-degree polynomials that cannot be factored further over the real numbers, typically of the form (x^2+bx+c). These factors lead to partial fractions with linear numerators.

Decomposition Method

For a rational function with irreducible quadratic factors, the decomposition includes terms where the numerator is a linear polynomial. If the denominator has an irreducible quadratic factor $(x^2 + bx + c)$,the decomposition is:

$$\frac{P(x)}{x^2 + bx + c} = \frac{Ax + B}{x^2 + bx + c}$$

Example: Decompose $\frac{2x^2+3x+4}{(x^2+1)}$

Solution:

Since the denominator is already an irreducible quadratic factor, set up the partial fractions:

$$\frac{2x^2 + 3x + 4}{x^2 + 1} = \frac{Ax + B}{x^2 + 1}$$

Clear the fractions by multiplying through by the denominator:

$$2x^2 + 3x + 4 = Ax + B$$

Solve for A and B by equating coefficients or substituting values of x:

Comparing coefficients of x and constants gives us:

$$A = 2$$

$$B = 3$$

Therefore, the decomposition is:

$$\frac{2x^2 + 3x + 4}{x^2 + 1} = \frac{2x + 3}{x^2 + 1}$$

6. WORKING EXAMPLES

1. Decompose $\frac{4x+7}{(x-1)(x+3)}$

Solution:

Set up the Partial Functions;

$$\frac{4x+7}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$$

Clear the fractions by multiplying through by the denominator:

$$4x+7 = A(x+3) + B(x-1)$$

Solve for A and B by substituting convenient values for x:

Let

$$x = 1: 4(1) + 7 = A(1+3) \Rightarrow 11 = 4A \Rightarrow A = \frac{11}{4}$$

Let

$$x = -3: 4(-3) + 7 = B(-3-1) \Rightarrow 5 = -4B \Rightarrow B = \frac{5}{4}$$

Therefore, the decomposition is:

$$\frac{4x+7}{(x-1)(x+3)} = \frac{11/4}{x-1} + \frac{5/4}{x+3}$$

2. Decompose $\frac{2x^2+5x+6}{(x+1)^2}$

Solution:

Set up the partial fractions:

$$\frac{2x^2+5x+6}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$$

Clear the fractions by multiplying through by the denominator:

$$2x^2+5x+6 = A(x+1) + B$$

Solve for A and B by substituting convenient values for x:

Let

$$x = -1: 2(-1)^2 + 5(-1) + 6 = B \Rightarrow 2 - 5 + 6 = B \Rightarrow B = 3$$

Differentiate both sides with respect to x and then substitute $x = -1$ $4x + 5 = A$

Let

$$x = -1: 4(-1) + 5 = A \Rightarrow A = 1$$

Therefore, the decomposition is:

$$\frac{2x^2 + 5x + 6}{(x+1)^2} = \frac{1}{x+1} + \frac{3}{(x+1)^2}$$

3. Decompose $\frac{3x^2+5x+2}{(x^2+4)}$

Solution:

Since the denominator is already an irreducible quadratic factor, set up the partial fractions:

$$\frac{3x^2 + 5x + 2}{x^2 + 4} = \frac{Ax + B}{x^2 + 4}$$

Clear the fractions by multiplying through by the denominator:

$$3x^2 + 5x + 2 = Ax + B$$

Solve for A and B by substituting convenient values for x:

Comparing the coefficients of x and constants gives us:

A = 3

B = 5

Therefore, the decomposition is:

$$\frac{3x^2 + 5x + 2}{x^2 + 4} = \frac{3x + 5}{x^2 + 4}$$

4. Decompose $\frac{2x^3+3x^2+4x+5}{(x-1)(x+1)^2}$

Solution:

$$\frac{2x^3 + 3x^2 + 4x + 5}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

$$2x^3 + 3x^2 + 4x + 5 = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$$

Expand and simplify

$$2x^3 + 3x^2 + 4x + 5 = A(x^2 + 2x + 1) + B(x^2 - 1) + C(x - 1)$$

$$2x^3 + 3x^2 + 4x + 5 = Ax^2 + 2Ax + A + Bx^2 - B + C_x - C$$

$$2x^3 + 3x^2 + 4x + 5 = (A + B)x^2 + (2A + C)x + (A - B - C)$$

Equate Coefficients:

Coefficient of x^2: $A + B = 3$

Coefficient of x: $2A + C = 4$

Constant term: $A - B - C = 5$

Solving the system of equations:

From $A + B = 3, B = 3 - A$

Substituting B into $A - B - C = 5$: $A - (3 - A) - C = 5 \Rightarrow 2A - 3 - C = 5 \Rightarrow 2A - C = 8$

Substitute C into $2A + C = 4$: $2A + 4 - 2A = 4 \Rightarrow C = 4 - 2A$

Therefore, $A = 2, B = 1, C = 0$

Therefore, the decomposition is:

$$\frac{2x^3 + 3x^2 + 4x + 5}{(x-1)(x+1)^2} = \frac{2}{x-1} + \frac{1}{x+1}$$

5. Decompose $\frac{7}{(x+1)(x+3)}$

Solution:

$$\frac{7}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$$

$$7 = A(x+3) + B(x+1)$$

Let

$$x = -1: 7 = A(-1+3) \Rightarrow 7 = 2A \Rightarrow A = \frac{7}{2}$$

Let

$$x = -3: 7 = B(-3+1) \Rightarrow 7 = -2B \Rightarrow B = -\frac{7}{2}$$

Therefore, the decomposition is:

$$\frac{7}{(x+1)(x+3)} = \frac{7/2}{x+1} - \frac{7/2}{x+3}$$

6. Decompose $\frac{x^2+4x+5}{(x+1)(x+2)}$

Solution:

$$\frac{x^2 + 4x + 5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$x^2 + 4x + 5 = A(x+2) + B(x+1)$$

$$x^2 + 4x + 5 = Ax + 2A + Bx + B$$

$$x^2 + 4x + 5 = (A+B)x + (2A+B)$$

$$A + B = 1$$

$$2A + B = 5$$

From $A + B = 1 \Rightarrow B = 1 - A$

$$2A + B = 5$$

$$2A + (1 - A) = 5$$

$$A = 4$$

$$B = 1 - 4 = -3$$

Therefore, the decomposition is:

$$\frac{x^2 + 4x + 5}{(x+1)(x+2)} = \frac{4}{x+1} - \frac{3}{x+2}$$

7. Decompose $\frac{2x+3}{(x-1)^2}$

Solution:

$$\frac{2x+3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$$

$$2x + 3 = A(x-1) + B$$

Let

$$x = 1: 2(1) + 3 = B \Rightarrow 5 = B$$

Differentiate both sides with respect to x and then substitute x = 1:

$$2 = A$$

Therefore, the decomposition is:

$$\frac{2x+3}{(x-1)^2} = \frac{2}{x-1} + \frac{5}{(x-1)^2}$$

8. Decompose $\frac{3x^2+7x+2}{(x+1)(x+3)}$

Solution:

$$\frac{3x^2 + 7x + 2}{(x+5)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$$

$$3x^2 + 7x + 2 = A(x+3) + B(x+1)$$

$$3x^2 + 7x + 2 = Ax + 3A + Bx + B$$

$$3x^2 + 7x + 2 = (A+B)x + (3A+B)$$

$$A + B = 3$$

$$3A + B = 2$$

$$B = 3 - A$$

Therefore,

$$3A + (3 - A) = 2$$
$$2A + 3 = 2$$
$$A = \frac{-1}{2}$$
$$B = 3 - \left(-\frac{1}{2}\right) = 3 + \frac{1}{2} = \frac{7}{2}$$

Therefore, the decomposition is:

$$\frac{3x^2 + 7x + 2}{(x+1)(x+3)} = \frac{-1/2}{x+1} + \frac{7/2}{x+3}$$

9. Decompose $\frac{x^2+x+1}{(x-1)(x+2)}$

Solution:

$$\frac{x^2 + x + 1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$
$$x^2 + x + 1 = A(x+2) + B(x-1)$$
$$x^2 + x + 1 = Ax + 2A + Bx - B$$
$$x^2 + x + 1 = (A+B)x + (2A - B)$$
$$A + B = 1$$
$$2A - B = 1$$
$$B = 1 - A$$

Therefore,

$$2A - (1 - A) = 1$$
$$3A - 1 = 1$$
$$3A = 2$$
$$A = \frac{3}{2}$$
$$B = 1 - \frac{2}{3} = \frac{1}{3}$$

Therefore, the decomposition is:

$$\frac{x^2 + x + 1}{(x-1)(x+2)} = \frac{2/3}{x-1} + \frac{1/3}{x+2}$$

10. Decompose $\frac{3x^3+5x^2+2x+4}{(x-1)^3}$

Solution:

$$\frac{3x^3+5x^2+2x^{+4}}{(x-1)^3} = \frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{(x-1)^3}$$

$$3x^3+5x^2+2x+4 = A(x-1)^2+B(x-1)+C$$

$$3x^3+5x^2+2x+4 = A(x^2-2x+1)+B(x-1)+C$$

$$3x^3+5x^2+2x+4 = Ax^2-2Ax+A+Bx-B+C$$

$$3x^3+5x^2+2x+4 = (A)x^2+(-2A+B)x+(A-B+C)$$

$$A = 3$$

$$-2A+B = 2$$

$$A-B+C = 4$$

$$-2(3)+B = 2$$

$$B = 8$$

$$3-8+C = 4$$

$$C = 9$$

Therefore, the decomposition is:

$$\frac{3x^3+5x^2+2x+4}{(x-1)^3} = \frac{3}{x-1}+\frac{8}{(x-1)^2}+\frac{9}{(x-1)^3}$$

11. Decompose $\frac{2x^4+3x^3+5x^2+7x+1}{(x^2+1)(x+2)}$

Solution:

$$\frac{2x^4+3x^3+5x^2+7x+1}{(x^2+1)(x+2)} = \frac{Ax+B}{x^{2+1}}+\frac{C}{x+2}$$

$$2x^4+3x^3+5x^2+7x+1 = (Ax+B)(x+2)+C(x^2+1)$$

$$2x^4+3x^3+5x^2+7x+1 = Ax^2+2Ax+Bx+2B+Cx^2+C$$

$$2x^4+3x^3+5x^2+7x+1 = (A+C)x^2+(2A+B)x+(2B+C)$$

$$2A+B = 3$$

$$2B+C = 1$$

$$A+C = 2$$

$$C = 2-A$$

$$2B+(2-A) = 1$$

$$B = \frac{A-1}{2}$$

$$2A+\frac{A-1}{2} = 3$$

$$A = \frac{7}{5}$$

$$B = \frac{\frac{7}{5} - 1}{2} = \frac{2}{10}$$

$$C = 2 - \frac{7}{5} = \frac{3}{5}$$

Therefore, the decomposition is:

$$\frac{2x^4 + 3x^3 + 5x^2 + 7x + 1}{(x^2 + 1)(x + 2)} = \frac{\frac{7}{5}x + 2/5}{x^2 + 1} + \frac{3/5}{x + 2}$$

12. Decompose $\frac{x^3+x^2+x+1}{(x-1)(x^2+x+1)}$

Solution:

$$\frac{x^3 + x^2 + x + 1}{(x - 1)(x^2 + x + 1)} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + x + 1}$$

$$x^2 + x^2 + x + 1 = A(x^2 + x + 1) + (Bx + C)(x - 1)$$

$$x^3 + x^2 + x + 1 = Ax^2 + Ax + A + Bx^2 - Bx + Cx - C$$

$$x^3 + x^2 + x + 1 = (A + B)x^2 + (A - B + C)x + (A - C)$$

$$A + B = 1$$

$$A - B + C = 1$$

$$A - C = 1$$

$$C = A - 1$$

$$A - B + (A - 1) = 1$$

$$2A - B = 2$$

$$2A - (1 - A) = 2$$

$$A = 1$$

$$B = 1 - 1 = 0$$

$$C = 1 - 1 = 0$$

Therefore, the decomposition is:

$$\frac{x^3 + x^2 + x + 1}{(x - 1)(x^2 + x + 1)} = \frac{1}{x - 1}$$

13. Decompose $\frac{x^4+2x^3+3x^2+4x+5}{(x^2+1)(x-2)^2}$

Solution:

$$\frac{x^4+2x^3+3x^2+4x+5}{(x^2+1)(x-2)^2}=\frac{Ax+B}{x^2+1}+\frac{C}{x-2}+\frac{D}{(x-2)^2}$$

$$x^4+2x^3+3x^2+4x+5=(Ax+B)(x-2)^2+C(x^2+1)(x-2)+D(x^2+1)$$

$$x^4+2x^3+3x^2+4x+5=(Ax+B)(x^2-4x+4)+C(x^3-2x^2+x-2)+D(x^2+1)$$

$$x^4+2x^3+3x^2+4x+5$$
$$=(Ax^3-4Ax^2+4Ax+Bx^2-4Bx+4B)+(Cx^3-2Cx^2+Cx-2C)$$
$$+Dx^2+D$$

$$x^4+2x^3+3x^2+4x+5$$
$$=Ax^3+Cx^3-4Ax^2-2Cx^2+Bx^2+Dx^2+4Ax-4Bx+Cx-2C+4B$$
$$+D$$

$$x^4+2x^3+3x^2+4x+5$$
$$=(A+C)x^3+(-4A-2C+B+D)x^2+(4A-4B+C)x$$
$$+(4B-2C+D)$$

$$A+C=1$$
$$C=1-A$$
$$-4A-2C+B+D=3$$
$$4A-4B+C=4$$
$$4B-2C+D=5$$
$$4A-4B+(1-A)=4$$
$$3A-4B=3$$
$$-4A-2(1-A)+B+D=3$$
$$-2A+B+D=5$$
$$4B-2(1-A)+D=5$$
$$4B+2A+D=7$$

From,

$$-2A+B+D=5$$
$$D=5+2A-B$$
$$4B+2A+(5+2A-B)=7$$
$$3B+4A=2$$

Multiply the $3A-4B=3$ by 3 and the $3B+4A=2$ by 4:

$$9A-12B=9$$

$$12B + 16A = 8$$

Add the two equations:

$$25A = 17 \Rightarrow A = \frac{17}{25}$$

Therefore,

$$3B + 4\left(\frac{17}{25}\right) = 2$$

$$B = -\frac{6}{25}$$

Find D using:

$$D = 5 + 2\left(\frac{17}{25}\right) - \left(-\frac{6}{25}\right)$$

$$D = 6 \cdot 6$$

Finally,

$$C = 1 - \frac{17}{25}$$

$$C = \frac{8}{25}$$

Therefore, the decomposition is:

$$\frac{x^4 + 2x^3 + 3x^2 + 4x + 5}{(x^2+1)(x-2)^2} = \frac{(17\,/\,25)x + \left(-{}^{6}\!/_{25}\right)}{x^2+1} + \frac{8\,/\,25}{x-2} + \frac{6 \cdot 6}{(x-2)^2}$$

7. SUMMARY

As we wrap up our unit on Rational Functions and Partial Fractions, let's take a moment to reflect on the critical insights and skills we've developed. Throughout this unit, we delved into the intricate world of rational functions, exploring their structures and how they can be simplified through the method of partial fractions. This technique not only enhances our algebraic proficiency but also significantly eases the process of integration in calculus.

Starting with the basics, we defined a rational function as a quotient of two polynomials. We distinguished between proper and improper rational functions based on the degrees of the numerator and the denominator. This distinction is crucial because it dictates the approach for partial fraction decomposition. For improper rational functions, we learned to perform polynomial division first, converting them into a sum of a polynomial and a proper rational function.

The decomposition of proper rational functions was particularly interesting. Depending on the factors of the denominator, whether distinct linear, repeated linear, or irreducible quadratic factors, we employed different strategies to break down these functions into simpler, more manageable parts. Each type of factor required a unique approach to determine the constants in the numerator of the partial fractions, enhancing our problem-solving toolkit.

One of the key aspects we focused on was the practical application of these techniques. By setting up equations from the partial fraction forms and solving for unknown constants, we practiced and perfected the method of clearing the denominators to simplify complex expressions. This skill is not only academically rewarding but also immensely useful in various fields of engineering, economics, and the natural sciences, where simplification of functions is frequently required.

Moreover, we tackled a variety of examples to reinforce our understanding and ensure mastery of the content. From simple linear factors to more complex quadratic factors, we explored how each scenario is approached and solved. This hands-on practice was vital in building confidence and competence in using partial fractions effectively.

As computer application students, the ability to manipulate and simplify rational functions using partial fractions is invaluable. It not only aids in computational tasks but also in developing algorithms that require optimization and efficient calculation methods. The logical thinking and analytical skills enhanced in this unit are transferable to software development and data analysis, where such mathematical tools often underpin complex programming functionalities.

This unit has equipped us with fundamental mathematical techniques that are pivotal in both academic and professional settings. The ability to decompose rational functions into partial fractions not only enriches our mathematical understanding but also prepares us for more advanced studies in calculus and beyond. As we move forward, let's continue to apply and expand upon these skills, exploring new challenges and applications in the fascinating world of mathematics and computer applications.

INSPIRED BY LIFE

8. GLOSSARY

- ***Rational Function:*** A function defined by the ratio of two polynomials, typically expressed as $\frac{P(x)}{Q(x)}$ where P(x) and Q(x) are polynomials
- ***Proper Rational Function:*** A rational function where the degree of the numerator polynomial P(x) is less than the degree of the denominator polynomial Q(x).
- ***Improper Rational Function:*** A rational function where the degree of the numerator polynomial P(x) is greater than or equal to the degree of the denominator polynomial Q(x).
- ***Partial Fractions:*** The expression of a rational function as a sum of simpler fractions, particularly used to simplify the integration process in calculus.
- ***Distinct Linear Factors:*** Terms in the denominator of a rational function that are linear (e.g., x−r) and do not repeat.
- ***Repeated Linear Factors:*** Linear terms in the denominator of a rational function that appear multiple times (e.g., $(x-r)^n$)
- ***Irreducible Quadratic Factors:*** Quadratic terms in the denominator of a rational function that cannot be factored into real linear factors (e.g. x^2+bx+c where the discriminant $b^2-4ac<0$)
- ***Polynomial Division:*** A method used to simplify an improper rational function into a sum of a polynomial and a proper rational function.
- ***Decomposition:*** The process of breaking down a proper rational function into partial fractions, each with simpler denominators.
- ***Constant Coefficients (A, B, C...):*** The numerators in the partial fractions, which are constants determined through algebraic methods to satisfy the equation formed by the original rational function.
- ***Factorization:*** The process of breaking down a polynomial into a product of simpler polynomials or factors, used in the partial fraction decomposition process.
- ***Integration:*** The mathematical process of finding the integral or antiderivative of a function, often facilitated by partial fractions in the case of rational functions.

9. SELF-ASSESSMENT QUESTIONS

1. A _____ rational function has a numerator with a lower degree than the denominator.
2. If the degree of the numerator is higher than the degree of the denominator, the rational function is called _____.
3. The process of expressing a rational function as a sum of simpler fractions is called _____.
4. _____ factors are those that appear more than once in the denominator of a rational function.
5. A rational function is proper if the degree of P(x) is _____ the degree of Q(x).
6. The method used to simplify an improper rational function into a sum of a polynomial and a proper rational function is called _____.
7. A term in the denominator that cannot be factored further over the real numbers is known as an _____ quadratic factor.
8. In partial fractions, constants such as A, B, and C in the numerators are determined by _____ both sides of the equation.
9. The denominator must be completely factored into linear and/or irreducible quadratic factors during the _____ step of partial fraction decomposition.
10. What is a rational function?
    A. A function that always equals zero.
    B. A function defined by the ratio of two polynomials.
    C. A function that cannot be integrated.
    D. A function that only contains integers.
11. What does it mean if a rational function is described as "improper"?
    A. The function is undefined.
    B. The numerator and denominator have the same degree.
    C. The degree of the numerator is less than the degree of the denominator.
    D. The degree of the numerator is greater than or equal to the degree of the denominator.
12. Which term describes the decomposition into simpler fractions?
    A. Factorization
    B. Polynomial division

C. Integration

D. Partial fractions

13. What are repeated linear factors?

A. Factors that cannot be simplified.

B. Factors that appear multiple times in the denominator.

C. Factors that only appear once.

D. Factors that are prime numbers.

14. What is necessary before setting up partial fractions?

A. Multiplying the denominator by the numerator.

B. Adding one to each factor of the denominator.

C. Factoring the denominator completely.

D. Simplifying the numerator.

15. What does polynomial division convert an improper rational function into?

A. A sum of two polynomials.

B. A polynomial plus a proper rational function.

C. Two distinct linear factors.

D. An irreducible quadratic function.

16. Which of the following is not a step in partial fraction decomposition?

A. Factoring the denominator.

B. Determining constants by setting x to zero.

C. Clearing the fractions by multiplying through by the denominator.

D. Solving for constants by equating coefficients.

17. In terms of partial fractions, what does a proper rational function look like after decomposition?

A. A single fraction.

B. A sum of multiple fractions with simpler denominators.

C. A single linear factor.

D. None of the above.

18. How are constants like A and B in the numerators of partial fractions typically found?

A. By trial and error.

B. By solving a differential equation.

C. By equating coefficients.

D. By integration.

19. Which type of factors might require a different approach in partial fraction decomposition?

A. Distinct linear factors.

B. Repeated linear factors.

C. Numerical coefficients.

D. Polynomial numerators.

20. What is typically the first step in the process of partial fraction decomposition?

A. Integrating the function.

B. Factoring the denominator completely.

C. Simplifying the numerator.

D. Applying the quotient rule.

10. TERMINAL QUESTIONS

1. Determine the rational function $\frac{5x^3+2x^2-x+1}{x^2-4x+3}$ is proper or improper
2. Decompose $\frac{6x+9}{x^2-9}$ into partial fractions
3. Simplify $\frac{x^2+3x+2}{x^3+x^2-x-1}$ using polynomial division
4. Factor the denominator and set up the partial fraction for $\frac{x^2-1}{(x-1)(x^2+x+1)}$
5. Calculate the constants for the partial fraction decomposition of $\frac{2x+3}{(x-1)(x+2)}$
6. Determine the derivative of the logarithmic function $log_b(x^3)$
7. Evaluate $\frac{3x^2+5x+2}{x^3+2x^2-x-2}$ at $x = 1$
8. Simplify $\frac{4x-5}{(x-2)^2}$ using partial fraction decomposition
9. Solve for x in the equation $\frac{x+1}{x^2+4x+4} = \frac{1}{x+2}$
10. Given $\frac{x^2-3x+2}{x^3-4x^2+5x-2}$, identify if the rational function is proper and explain why.
11. Decompose $\frac{2x+3}{x^2+x-6}$ into partial fractions.
12. Find the constants A and B in $\frac{5x+7}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$
13. Derive the formula for the derivative of $log_a(x)$
14. Use partial fractions to integrate $\frac{2x-1}{x^2-4}$
15. Solve for x where $\frac{4x^2+5x+1}{x+1} = 0$
16. Decompose $\frac{x^3+2}{x^2-x-2}$ using partial fraction decomposition.
17. Factor $x^2 + 5x + 6$ and use this factorization to decompose the corresponding rational function.

11. ANSWERS

Self-Assessment Questions

1. Proper
2. Improper
3. Partial fraction decomposition
4. Repeated linear
5. less than
6. Polynomial division
7. Irreducible quadratic
8. Multiplying
9. Factoring
10. B) A function defined by the ratio of two polynomials.
11. D) The degree of the numerator is greater than or equal to the degree of the denominator.
12. D) Partial fractions
13. B) Factors that appear multiple times in the denominator.
14. C) Factoring the denominator completely.
15. B) A polynomial plus a proper rational function.
16. B) Determining constants by setting x to zero.
17. B) A sum of multiple fractions with simpler denominators.
18. C) By equating coefficients.
19. B) Repeated linear factors.
20. B) Factoring the denominator completely.

Terminal Questions

1. Proper, because the degree of the numerator is greater than the degree of the denominator.
2. $\frac{6x+9}{x^2-9} = \frac{A}{x-3} + \frac{B}{x+3}$ where A = 3, B = 3
3. $\frac{x-1}{x+1} + \frac{1}{(x+1)^2}$.

4. Set up as: $\frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$
5. A = 1, B = 1
6. $\frac{3}{x\,ln(b)}$.
7. Value at x = 1 is 10
8. A = -3, B = 4
9. x = 0
10. Proper, because the degree of the numerator is less than that of the denominator.
11. A = 1, B = 1
12. A = 4, B = 1
13. $\frac{1}{x\,ln(a)}$.
14. $ln\left|\frac{x-2}{x+2}\right|$.
15. $x = -\frac{1}{4}$.
16. A = 1, B = 1, C = 1
17. Factors are: $(x+2)(x+3)$, A = -1, B =1